Do numeric anchors have an effect on estimates of math equations?

IB Psychology HL Internal Assessment

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Introduction

Within the cognitive perspective of psychology, many researchers theorize potential explanations for how humans think. One such theory is the Priming Theory, proposed by Iyengar and colleagues in 1982. This theory states that external stimuli may affect human responses by activating mental constructs without them consciously realizing it (Weingarten, E., Chen, Q., McAdams, M., Yi, J., Hepler, J., & Albarracín, D., 2016). A potential consequence of the priming theory is the "anchoring effect" (Valdez, A. C., Ziefle, M., & Sedlmair, M., 2018), which describes the tendency of individuals to make judgments biased toward the first piece of information received (O'Garra, T., & Sisco, M. R., 2020). This information is called an "anchor" and is an example of environmental stimuli that can influence responses. This is relevant as the present investigation will investigate the role of anchors on problem-solving.

The present investigation is a simplified replication of Study One of "Biased Calculations: Numeric Anchors Influence Answers to Math Equations," conducted by Smith and Windschitl in 2011[1]. The researchers presented participants with a series of mathematical addition equations, along with an anchor value, which was either high or low relative to the actual solution. The researchers found that, based on the participants' answers within a time constraint, their estimates tended toward the anchor value. Based on their findings, the present investigation aims to test whether comparisons with irrelevant anchors in high school students influence estimates based on calculations. This is worth studying because anchoring biases can influence human behavior and decision-making, so learning its potency can allow humans to see its importance and, thus, how to reduce its impact. For example, a grocery store shopper trying to

[1] Full published journal can be found at journal.sjdm.org/index.htm; more information included in references.

calculate the total cost of his groceries might see that a new gas grill (an item he is not buying) costs $199.99 (Smith, A., & Windschitl, P., 2011), and this might influence the shopper's estimate of his grocery contents to be biased toward the cost of the grill.

The independent variables in this study are the anchor values the researchers will provide to the participants and the equations, which will either be "high" compared to the equation's solution ($2100 \leq X \leq 2300$, where 'X' represents the anchor value) or "low" ($700 \leq X \leq 900$). The dependent variable is the estimation difference from the actual solution. This relates to the anchoring effect as the unrelated number (anchor) the participants will receive may bias their estimate of the equation toward that number.

H_0: If irrelevant anchors do not influence estimation, then when presented with an equation, the average of the participants' estimates will be within 100 units of the solution regardless of the condition.

H_1: If irrelevant anchors influence estimation, then, when presented with an equation, the average of the participants' estimates will be at least 100 units greater than the solution if in the "high anchor" condition and at least 100 units less than the solution if in the "low anchor" condition.

Exploration

Research Design

The experiment used a repeated measures design, in which multiple measures are used on each participant (Sullivan, L., 2008). This design was chosen to allow every participant to go through both a high and low anchor condition for correlations between anchors and estimates to be made. Participants were given two equations: one alongside a high anchor, and the other alongside a low anchor.

Sample

The participants were selected through convenience sampling, which draws participants from a source easily accessible to the researcher(s) (Andrade, C., 2021). It was chosen because we, as the researchers, are high school students, so selecting other high school students as our participants was the most convenient option. Additionally, we aim to test the anchoring effect in high school students specifically. No participant selected was aware of the study. Additionally, all participants were over the age of consent, as they were juniors and seniors aged 16 to 18, and currently take at least one math course.

Controls

For controls, participants were read standardized instructions (see Appendix 2) to ensure each received the same instruction and to reduce the potential bias of researcher influence. Additionally, participants were placed in different rooms: one for selection, one for testing, and one after they finished. This would prevent a participant bias in which certain participants explain the experiment to others who have yet to participate.

Materials

- Pens for writing utensils (prevented changing answers)
- Random number/addition equation generator (prevented researcher bias and saved time) (Appendix 13)
    - $1100 \leq Y \leq 1900$, where Y is the solution of the equation
    - Minimum value for addends: 367
    - Maximum value for addends: 633
- Index cards (participants could write down their answers; small enough to prevent them from solving the equations by hand)
- Timer
- Google Spreadsheet (used for data collection)
- Google Slideshow (presented anchors and equations to participants)
- Chromebook (used to present the slideshow)
- Internal and external scripts (Appendices 11 and 2)
    - Used for the researchers to know the procedure and for the participants to be instructed

Procedure

- Researchers A and C took half of the participants for testing one-by-one (10), and researchers B and D took the other half to save time
- Each participant was told to sit down and read/sign the consent form (Appendix 1)
- Participants were introduced to study and given instructions

- Researchers B/D presented the Google Slideshow with the anchors and equations (see Appendix 4 for anchors and equations) and would switch after multiple intervals
- Researchers A/C read the external script verbatim (Appendix 2) and held a timer to let researchers B/D know when to switch slides
- Participants were asked to look at a number (the anchor) for 3 seconds
- Researchers B/D then switched slides and researchers A/C asked the participants if the solution to the equation was higher or lower than the number
- Participants were then instructed to write down their answer to that question
- Participants were then asked to estimate the solution within 10 seconds, and wrote it down
- Repeated for each participant (high and low anchor condition)
- After the study was completed, participants were thanked and debriefed via email (Appendix 12)

Analysis

Some of the participants' answers were outliers, thus, their answers were removed as they altered our results (see Appendix 10 for justification of removal). These may have happened due to typos, not following instructions, or not paying attention to the equations.

Descriptive Statistics

Figure 1. *Mean, Range, Variance, and Standard Deviation of Distance from Solution for both conditions (outliers excluded).*

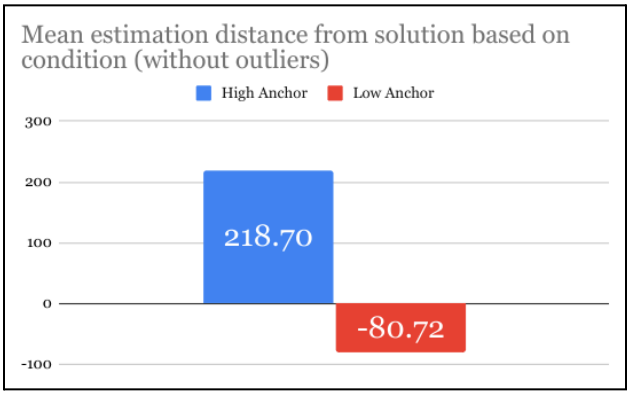
WITHOUT OUTLIERS	**Condition 1 (High Anchor, HA)**	**Condition 2 (Low Anchor, LA)**
Mean	218.70	-80.72
Range	1157.00	793.00
Variance	164733.87	26840.57
Standard Deviation	405.87	163.83

On average, the HA condition's estimates were 218.70 units greater than the solution in comparison to -80.72 units in the LA condition, indicating that the estimates were affected in the positive or negative direction based on the anchor being high/low relative to the solutions. The range in the HA condition (1157.00) was 364 units greater than that of the LA condition (793.00), indicating inconsistency, making it unreliable because there were numerous skews. The variance further explains the inconsistency of the HA condition with it being over six times greater (164733.87) than that of the LA condition (26840.57). It also means that the HA condition's results are less reliable than the LA condition's. The standard deviation of the HA condition (405.87) also demonstrates inconsistency compared to the LA condition (163.83),

being 242.04 units greater (see Appendix 6 for descriptive statistics including outliers). The range, variance and standard deviation values of the two conditions suggest that lower anchors may not affect subsequent responses as much as higher anchors may, as well as inconsistencies in the data overall reducing the reliability of the results.

Graph

Figure 2.



As shown in Figure 2, the LA condition had a lower mean estimate distance from the solution (-80.72) versus the HA condition (218.70). While the HA condition did support our research hypothesis as the participants' average estimates were at least 100 units greater than the solution (218.70), the LA condition did not quite support our hypothesis as the participants' average estimates were not at least 100 units less than the solution (-80.72). Therefore, our research hypothesis can be rejected and our null hypothesis must be retained.

Inferential Statistics

Figure 3.

Spearman's Rho Test		
Condition	**R-value**	**P-value (two-tailed)**
High Anchor w/o outliers	0.45247	0.05176
Low Anchor w/o outliers	0.03522	0.88967

Without outliers, the participants' estimates were closer to the actual solution (80.72 units away) in the LA condition than in the HA condition (218.70 units away). A Spearman's Rho test indicated that the difference between the estimates in both the HA and LA conditions and the actual solutions were statistically insignificant, as $p > 0.05$ in both conditions. Additionally, the r-value in each of these conditions were both less than 0.7, meaning the relationship between the anchor provided and the estimates yielded was weak (see Appendix 7 for Spearman's Rho Test including outliers).

However, it can be deduced that there was a more significant effect of the anchors provided on the estimates yielded in the HA condition than in the LA condition, as the p-value was about 0.052 in the HA condition compared to ≈0.89 in the LA condition. Also, the r-value in the HA condition (≈0.41) was much higher than in the LA condition (≈0.04), suggesting a stronger correlation between the anchors and estimates in the HA condition.

This backs up our decision to reject our research hypothesis because neither of the conditions displayed a statistically significant relationship between the anchors provided and the estimates yielded. Although we did not explicitly mention in our hypothesis that there would be a

statistically significant relationship between the two, our predictions between the relationships of the anchors and estimates in both conditions calls for some significance.

Evaluation

The Priming Theory predicts that environmental stimuli may affect subsequent responses as individuals tend to make judgments that are biased toward the first piece of information they receive. In the context of mathematics, this may mean that an individual's estimate of an equation will be biased toward an anchor value provided. In the aforementioned study on the Priming Theory, they found that their anchor values had a statistically significant impact on the estimates yielded from the participants ($p < .001$). However, in our replication of the study, the closest we got to a statistically significant effect of the anchors on the estimates was in the HA group without outliers, where $p < .06$, meaning this was statistically insignificant. This may be due to the change in the time limit, or because each participant is a current high school student, and thus their mathematical skills are fresher than the average human used in the original study. This means that our results do not support the theory.

One **strength** of using a repeated measures design for our experiment is that it controlled the confounding variable of varying estimation skills. This means we were able to avoid the possible effects that a differing amount of estimation skill would have on our results. One **limitation** is that it allowed for the participants to have experience of the experiment the second time around, formally known as the carryover effect. This may have increased accuracy in the low anchor condition versus the HA condition (supported by a lower mean estimation distance in the LA condition than the HA condition).

One **strength** of our opportunity sample is that it allowed our participants to be gathered easily from the same environment, being representative of our target population (high school students) as they were all juniors and seniors. One **limitation** is that some researchers knew some of the participants, which possibly affected their comfortability/effort. This may have skewed the results to either be more accurate or inaccurate due to demand characteristics.

One **strength** of our procedure/materials is that we used an external script (see Appendix 2) to provide experimental instructions to each of the participants. This reduced the amount of variance in instruction provided to each participant, eliminating some researcher bias. One **limitation** is that the external script did not address participants having extra time which was unintentionally added onto their total amount of time to estimate, creating a methodological bias. This could have possibly skewed the results and made the participants more accurate, reducing the chance for the Priming Theory to have a significant impact.

One **modification** that could have been made would be to use counterbalancing, a procedure that allows researchers to control the effects of nuisance variables in designs where the same participants are subjected to numerous conditions or stimuli (Corriero, E., 2017). For our experiment specifically, this could have been done by having half of the participants go through the HA condition first and the second half going through the LA condition first, potentially reducing the bias created by the participants going through a repeated measures design. Another **modification** is that we could have included privacy shields in front of the participants during the duration of the testing. This could have prevented the amount of possible demand

characteristics as participants would be less likely to change their effort without being able to see the researchers actively during the experiment. A third **modification** is that we could have included a time limit to the higher/lower question separate from the time limit used when asked to estimate the solution to the equations. This would have prevented the participants from having extra time to estimate the solution than intended, and may have displayed more of a reliance on the anchor value for their estimate.

Overall, we conclude that because our study was not significant at $p < 0.06$ and $p < 0.9$ that the anchors did not have an impact on the estimates, suggesting that the Priming Theory does not have a significant impact. However, this could be due to the nature of the estimation process being semi-random regardless of information provided.

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Appendices

Appendix 1. Letter of Consent Form

Letter of Consent

Hello, we are performing an experiment as part of our internal assessment for our IB Psychology class. We are investigating **numeric estimations**.

We would like to ask you to be a part of our experiment.

If you agree to take part in this experiment, you should know that:

- All data that we obtain will be kept confidential and anonymous
- You may stop participating in this experiment at any time.
- You shall receive information about the nature of the experiment and our results after we have obtained the results.

The experiment will take about **3** minutes to complete.

I, _____, understand the nature of this experiment and I agree to participate voluntarily. I give the researchers permission to use my data as a part of their experimental study.

Signature: _____ Date(s): **10/26/23, 10/30/23**

Appendix 2. External Script

Roles: Speakers - EXP A, EXP B; Slides - EXP C, EXP D

General Introduction:

EXP A - "Good morning, everyone! Today, my colleagues and I have an experiment for you all to take part in. First, we are going to split the class into two groups. Each of you will come up and take a piece of paper out of this bag. If you get a paper with the number '1' on it, you will be with EXP A and EXP C, just outside of the library. If you get a paper with the number '2' on it, you will be with EXP B and EXP D, just outside of [redacted]'s room. After the experiment, you will come back to [redacted]'s room. Any questions? (they ask questions if they have any and we answer) Alright, you all can come up." (groups are sorted and transported to Library / [redacted]'s room)

In the Library / [redacted]'s Room:

"Alright, who would like to go **first/next**? (someone is selected) Alright, follow us. (goes to station)"

(EXP#A/B gives consent form) "Here is our consent form. Please read it, then sign your name and write today's date." (they sign)

Introduction to Experiment:

"Welcome to our psychology experiment. Today, you are going to take part in a 'math estimation game'. First, you will be given a random number and then asked if the solution to the following equation is higher or lower than the given number, and you will estimate the solution to the equation. Then, you will be given another random number and then asked if the solution to the following equation is higher or lower than the previous number, and you will again estimate the solution to the equation. On the index card in front of you, write your answer to both questions; you should have two answers for both equation #1 and equation #2. We ask you to try to be as accurate as possible. Do you have any questions? (they ask questions if they have any and we answer)"

Experiment (both conditions):

"First, I am going to present one number for the first equation, and you will have 3 seconds to look at it. After 3 seconds, I will present an equation for 10 seconds. Ready? (they say yes) Okay. Look at this number. (EXP C/EXP D presents anchor for 3 seconds, then switches to equation). Is the solution to this equation higher or lower than that number? (they say higher/lower).

Alright, write that down. (they write it down) And what is your estimate for this question? You have 10 seconds to answer. When you finish, write down your answer. (10 seconds passes, exp moves to the next anchor). Look at this number. (present anchor for 3 seconds, switch to equation). Is the solution to this equation higher or lower than that number? (they say higher/lower). Alright, write that down. (they write it down) And what is your estimate for this question? You have 10 seconds. (10 seconds passes). Alright! Thank you for participating!"

Appendix 3. Data collection tables

	Condition One					
Participant #	Anchor	Equation	Solution	Estimate	Higher/Lower	Distance
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

	Condition Two					
Participant #	Anchor	Equation	Solution	Estimate	Higher/Lower	Distance
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

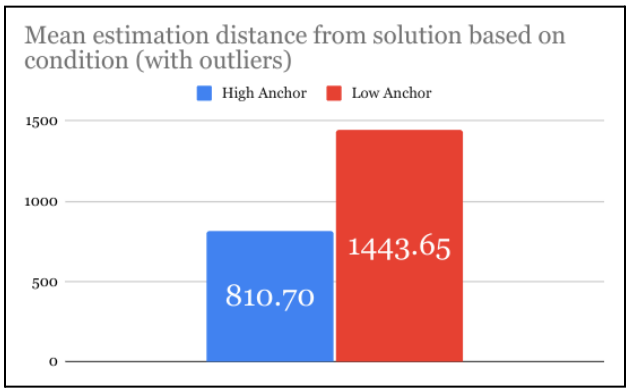
Appendix 4. Raw data

	Condition One					
Participant #	Anchor	Equation	Solution	Estimate	Higher/Lower	Distance
1	2153	388 + 413 + 500	1301	1230	Higher	-71
2	2221	416 + 583 + 538	1537	2100	Lower	563
3	2109	457 + 517 + 445	1419	1452	Lower	33
4	2146	482 + 435 + 507	1424	1600	Lower	176
5	2200	628 + 544 + 370	1542	1700	Lower	158
6	2265	543 + 590 + 491	1624	3000	Higher	1376
7	2231	592 + 405 + 425	1422	2100	Lower	678
8	2170	443 + 430 + 516	1389	1280	Higher	-109
9	2204	579 + 423 + 440	1442	13500*	Higher	12058
10	2274	630 + 443 + 464	1537	2262	Lower	725
11	2268	564 + 565 + 600	1729	1725	Lower	-4
12	2220	567 + 516 + 462	1545	1523	Lower	-22
13	2244	595 + 520 + 515	1630	2100	Lower	470
14	2258	573 + 376 + 470	1419	1200	Higher	-219
15	2106	453 + 617 + 458	1528	1534	Higher	6
16	2132	542 + 605 + 478	1625	2133	Higher	508
17	2107	591 + 600 + 559	1750	1600	Lower	-150
18	2262	523 + 468 + 526	1517	1650	Higher	133
19	2215	548 + 477 + 495	1520	1600	Higher	80
20	2174	503 + 408 + 564	1475	1300	Higher	-175

	Condition Two					
Participant #	Anchor	Equation	Solution	Estimate	Higher/Lower	Distance
1	855	488 + 502 + 478	1468	1370	Lower	-98
2	711	409 + 504 + 399	1312	1200	Higher	-112
3	877	525 + 432 + 526	1483	1256	Higher	-227
4	798	475 + 597 + 461	1533	1500	Higher	-33
5	796	484 + 484 + 464	1432	1300	Higher	-132
6	836	612 + 452 + 476	1540	14000*	Higher	12460
7	851	373 + 500 + 458	1331	1398	Higher	67
8	731	606 + 471 + 489	1566	1510	Higher	-56
9	840	482 + 540 + 612	1634	19500*	Higher	17866
10	794	385 + 462 + 398	1245	1638	Higher	393
11	775	543 + 438 + 397	1378	1400	Higher	22
12	812	589 + 426 + 577	1592	1476	Higher	-116
13	755	500 + 612 + 398	1510	1400	Higher	-110
14	867	476 + 475 + 470	1421	1200	Lower	-221
15	825	387 + 401 + 480	1268	1200	Higher	-68
16	806	485 + 517 + 440	1442	1232	Higher	-210
17	813	550 + 547 + 586	1683	1500	Lower	-183
18	851	558 + 623 + 633	1814	1800	Lower	-14
19	836	512 + 468 + 575	1555	1600	Lower	45
20	704	576 + 457 + 495	1528	1128	Lower	-400

* Outliers.

Appendix 5. Estimation Graph with Outliers



Appendix 6. Mean, Range, Variance, and Standard Deviation for both conditions (outliers included)

WITH OUTLIERS	**Condition 1 (High Anchor)**	**Condition 2 (Low Anchor)**
Mean	810.70	1443.65
Range	12277.00	18266.00
Variance	7164471.27	22807190.98
Standard Deviation	2676.65	4775.69

Appendix 7. Spearman's Rho Test (outliers included)

Spearman's Rho Test		
Condition	**R-value**	**P-value (two-tailed)**
High Anchor w/outliers	0.41328	0.07011
Low Anchor w/outliers	0.13424	0.57259

Appendix 8. Spearman's Rho Test Input/Output Values (outliers included)

High Anchor Condition

Spearman's Rho Calculator

The value of r_s is: 0.41328.

X Values	Y Values
2153	1230
2221	2100
2109	1452
2146	1600
2200	1700
2265	3000
2231	2100
2170	1280
2204	13500
2274	2262
2268	1725
2220	1523
2244	2100
2258	1200
2106	1534
2132	2133
2107	1600
2262	1650
2215	1600
2174	1300

r_s = 0.41328, p (2-tailed) = 0.07011.

Low Anchor Condition

Spearman's Rho Calculator

The value of r_s is: 0.13424.

X Values	Y Values
855	1370
711	1200
877	1256
798	1500
796	1300
836	14000
851	1398
731	1510
840	19500
794	1638
775	1400
812	1476
755	1400
867	1200
825	1200
806	1232
813	1500
851	1800
836	1600
704	1128

r_s = 0.13424, p (2-tailed) = 0.57259.

Appendix 9. Spearman's Rho Test Input/Output Values (outliers excluded)

High Anchor Condition

Spearman's Rho Calculator

The value of r_s is: 0.45247.

X Values	Y Values
2153	1230
2221	2100
2109	1452
2146	1600
2200	1700
2265	3000
2231	2100
2170	1280
2274	2262
2268	1725
2220	1523
2244	2100
2258	1200
2106	1534
2132	2133
2107	1600
2262	1650
2215	1600
2174	1300

r_s = 0.45247, p (2-tailed) = 0.05176.

Low Anchor Condition

Spearman's Rho Calculator

The value of r_s is: 0.03522.

X Values	Y Values
855	1370
711	1200
877	1256
798	1500
796	1300
851	1398
731	1510
794	1638
775	1400
812	1476
755	1400
867	1200
825	1200
806	1232
813	1500
851	1800
836	1600
704	1128

r_s = 0.03522, p (2-tailed) = 0.88967.

Appendix 10. Justification for Outliers

Condition 1 (High Anchor) Estimates:

1200	1230	1280	1300	1452	1523	1534	1600	1600	1600
1650	1700	1725	2100	2100	2100	2133	2262	3000	13500

Median = (1600 + 1650) / 2 = 1625

IQR = Q3 - Q1
Q1 = 1452
Q3 = 2100
IQR = 2100 - 1452 = 648

High Outliers ≥ Q3 + (1.5 * IQR) ≥ 2100 + (1.5 * 648) ≥ 3072
Low Outliers ≤ Q1 - (1.5 * IQR) ≤ 1452 - (1.5 * 648) ≤ 480

High Outliers: 13500
Low Outliers: None

Condition 2 (Low Anchor) Estimates:

1128	1200	1200	1200	1232	1256	1300	1370	1398	1400
1400	1476	1500	1500	1510	1600	1638	1800	14000	19500

Median = (1400 + 1400) / 2 = 1400

IQR = Q3 - Q1
Q1 = 1232
Q3 = 1600
IQR = 1600 - 1232 = 368

High Outliers ≥ Q3 + (1.5 * IQR) ≥ 1600 + (1.5 * 368) ≥ 2152
Low Outliers ≤ Q1 - (1.5 * IQR) ≤ 1232 - (1.5 * 368) ≤ 680

High Outliers: 14000, 19500
Low Outliers: None

Appendix 11. Internal Script

- Before Study
    - EXP#B: Generate 20 equations $X_1 + X_2 + X_3 = Y$ where values of X are $367 \leq X \leq 633$ and value of Y is $1100 \leq Y \leq 1900$. Generate anchor between values of $2100 \leq Z \leq 2300$; repeat but for $700 \leq Z \leq 900$
    - EXP#A/B: Type all equations and anchors into Google Sheets
- Introduce study/experiment to participant (one at a time)
- Condition One: High Anchor, 10 sec. Limit
    - EXP#A/C/D: Explain steps of the experiment to participants
    - EXP#A/C/D: Present anchor first for 3 seconds, before presenting the equation. Ask the participant if the anchor is less than or more than the solution to the equation. Ask what their estimate of the solution is (written on a piece of paper in pen); 10 second time limit for estimate
- Condition Two: Low Anchor, 10 sec. limit
    - EXP#A/C/D: Explain steps of the experiment to participants
    - EXP#A/C/D: Present anchor first for 3 seconds, before presenting the equation. Ask the participant if the anchor is less than or more than the solution to the equation. Ask what their estimate of the solution is (written on a piece of paper in pen); 10 second time limit for estimate
- Data Collection:
    - EXP#B: After equations are calculated by each participant, EXP#B will put all results into a Google Sheet. Analyzations will be made by each EXP# for each condition, and EXP# will draw a final conclusion about results, as well as comparing results to the original hypothesis.

Appendix 12. Debriefing Email

Good Morning,

Thank you so much for participating in our experiment for our IB Psychology Internal Assessment earlier this year. Our experiment was a replication of the Smith and Windschitl (2011) study and aimed to test priming theory. There is a link to the original study below. This theory or model states that people tend to rely on the first piece of information given when making decisions. In our case, we specifically tested the effects of an "anchor" value (a number given alongside an equation). Our results ultimately found that there was a relationship present between the anchors and estimations, however, the results showed that this relationship was unclear. If you have any additional questions about our study, the results or what the results will be used for, please let us know.

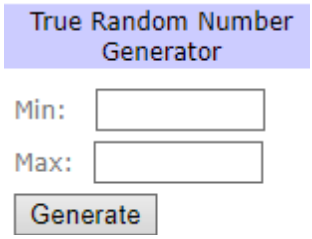
Link to study:
https://www.cambridge.org/core/journals/judgment-and-decision-making/article/biased-calculations-numeric-anchors-influence-answers-to-math-equations/939E75AF2B4984C7A56C9E2FBAD444DA

Thank you again for your help,

[redacted]

Appendix 13. Random Number/Addition Generator

Random Number Generator:
https://www.random.org



Random Addition Equation Generator -
https://www.convertsimple.com/generate-addition-equation/

